

Checkout System Improvement

Portfolio Case Study (Simulated)

Prepared by: Ammar Ali

1. Project Overview

This case study focuses on an e-commerce checkout process, where a user selects items, reviews the cart, proceeds to payment, and receives order confirmation. The goal is to improve the reliability and usability of the checkout flow to reduce cart abandonment and improve conversion.

2. Problem Statement

The checkout experience is currently impacted by two critical issues:

- Checkout button is unresponsive, preventing users from proceeding to payment.
- Selected items fail to appear on the cart display page, causing users to lose confidence in the order total.

These issues are contributing to measurable business impact, including:

- Approximately 20% increase in cart abandonment.
- Approximately 10% increase in customer churn.
- Approximately 20% increase in checkout-related support tickets.

3. Scope

3.1 In Scope

- Cart page reliability (accurate item display, pricing, quantities, and totals).
- Checkout initiation (responsive checkout button and redirect to payment gateway).
- Payment authorization handling (approved/declined response handling).
- User feedback (confirmation messaging, error messaging, and email receipt on success).

3.2 Out of Scope

- Search functionality and browsing experience.
- Product recommendations or “similar items”.
- Stored payment methods / saved cards.
- Order tracking page redesign or loyalty programs.

4. Stakeholders

Stakeholder	Type	Goal	Current Pain	Influence
Customer	External	Complete a purchase smoothly and confidently	Cart items missing and checkout failures prevent purchase completion	Medium
Business Owner	Internal	Increase revenue and reduce customer churn	Increased cart abandonment and lost conversions leading to revenue impact	High
Product Manager	Internal	Improve checkout conversion and reduce friction in the purchase flow	Checkout bottlenecks, user complaints, and drop offs during cart/checkout	High
Developers	Internal	Implement a stable fix based on clear testable requirements	Unclear requirements and incomplete issue context leading to rework and debugging	High
Customer Support	Internal	Resolve checkout issues quickly and reduce ticket volume	Increased volume of checkout related support tickets	Medium
Payment Gateway Provider	External	Process authorization requests reliably and consistently	Failed authorization requests and integration related errors	Medium

5. Current State Analysis

5.1 Current Process Flow

The user selects items and adds them to the cart. The system displays cart contents on the cart page, where the user can review items and attempt to proceed by selecting the checkout button. When checkout is selected, the system is expected to redirect the user to the payment gateway interface and submit a payment authorization request. The payment gateway returns an approval or decline response, which the system displays to the user. However, the checkout flow does not execute reliably due to the issues described below.

5.2 Current Issues Identified

- Cart page fails to display all items selected by the user.
- Checkout button is unresponsive, preventing redirect to the payment gateway.
- Payment feedback and error messaging are inconsistent or unclear when a transaction cannot be completed.

5.3 Business Impact

Checkout failures are increasing abandoned purchases and driving additional customer support workload. Users are leaving the purchase flow due to cart display inconsistencies and an unreliable checkout redirect, resulting in lost conversions and increased customer dissatisfaction.

6. Future State Analysis

6.1 Future Process Summary

The system shall consistently display all items added to the cart, including unit price, quantity, per-item subtotal, and cart total. The checkout button shall remain responsive and redirect the user to the payment gateway interface when selected. The system shall submit payment authorization requests to the payment gateway and display the returned authorization status (approved/declined) to the user. If approved, the system shall display an order confirmation and send an electronic receipt to the user's email. If declined, the system shall display a clear, actionable error message to guide the user to correct the issue and retry payment.

6.2 Expected Outcomes

- Up to 20% reduction in cart abandonment (returning toward baseline levels).
- Up to 10% reduction in customer churn.
- Up to 20% reduction in checkout-related support tickets.

7. Assumptions and Risks

7.1 Assumptions

- Checkout issues are reproducible and attributable to the cart/checkout workflow (not inventory availability).
- Payment gateway returns standard authorization statuses (approved/declined) and error codes on declines.
- User email is available at checkout (guest checkout collects email or user is logged in).

7.2 Risks

- Payment gateway latency or timeouts during peak traffic could impact redirect and authorization performance.
- Cart state inconsistencies across sessions/devices may cause display failures.
- Regression risk: checkout fixes may affect pricing/total calculations and require thorough testing.

8. Process Flow

8.1 Process Flow Diagram

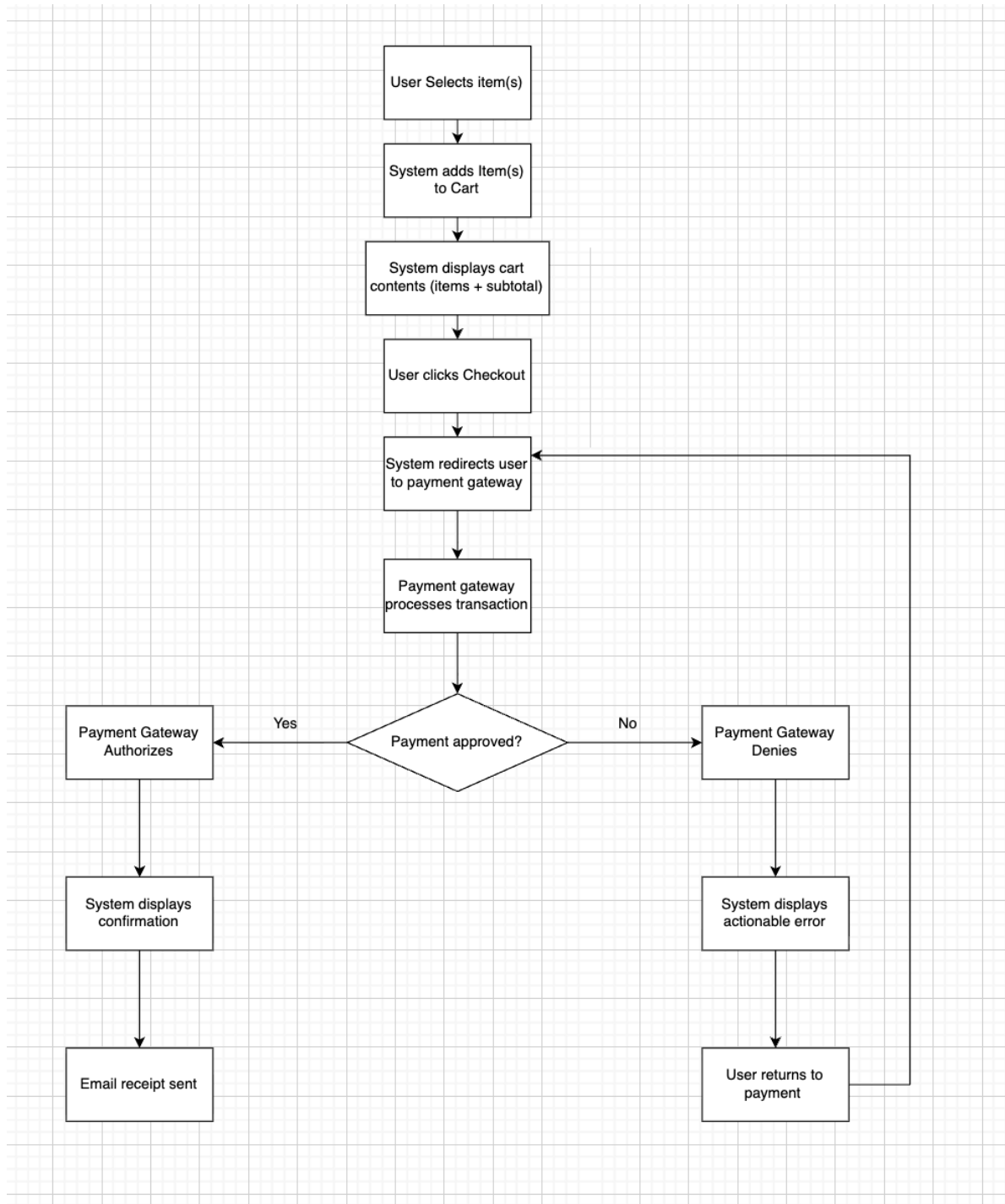


Figure 1. Checkout process flow (improved).

8.2 System Interaction Overview

The user adds items to the cart and selects the cart button. The system displays the cart contents, including selected items and the subtotal. The user selects the checkout button, and the system sends a payment

authorization request to the external payment gateway. The payment gateway returns an approval or decline response. If approved, the system displays an order confirmation to the user and sends an electronic receipt to the user's email. If declined, the system displays a clear, actionable error message to guide the user to correct the issue and retry payment.

8.3 Key Data Elements

Cart Data

- Item ID
- Unit price
- Quantity
- Subtotal
- Cart total

Payment Data

- Payment amount
- Authorization status (approved/declined)
- Decline error code

Order / Confirmation Data

- Order ID
- Transaction ID
- Transaction timestamp
- Receipt email status (sent/failed)

9. Requirements

9.1 Functional Requirements

ID	Requirement
FR-01	The system shall display all items added to the cart by the user on the cart page.
FR-02	The system shall display the correct unit price and quantity for each cart item.
FR-03	The system shall calculate and display the per-item subtotal and the total cart amount.
FR-04	The system shall allow the user to remove items from the cart.
FR-05	The system shall allow the user to update item quantities in the cart.
FR-06	The system shall allow the user to proceed to checkout when the checkout button is selected.
FR-07	The system shall redirect the user to the payment gateway interface upon checkout selection.

FR-08	The system shall securely transmit submitted payment details to the payment gateway for authorization.
FR-09	The system shall display the payment authorization status (approved/declined) returned by the payment gateway.
FR-10	The system shall display an order confirmation and send an electronic receipt to the user's email upon payment approval; if declined, the system shall display a clear, actionable error message.

9.2 Non-Functional Requirements

ID	Requirement
NFR-01	The system shall complete checkout page loading and payment redirection within 3 seconds under normal operating conditions.
NFR-02	The system shall generate and send the electronic receipt within 60 seconds of successful payment authorization.
NFR-03	The system shall support at least 5,000 concurrent checkout sessions without performance degradation.
NFR-04	The system shall encrypt all payment data using industry-standard encryption protocols (e.g., TLS 1.2 or higher).
NFR-05	The system shall log all successful and failed checkout transactions for auditing and reporting purposes.

9.3 Acceptance Criteria

AC-01 Cart Display

- Given the user has added items to the cart
- When the user selects the cart button
- Then the system displays all cart items with the correct name, price, and quantity.

AC-02 Checkout Redirect

- Given the user has items in the cart
- When the user selects the checkout button
- Then the system redirects the user to the payment gateway interface.

AC-03 Payment Approval

- Given the user enters valid payment details
- When the payment gateway approves the transaction
- Then the system displays an order confirmation and sends an electronic receipt to the user's email.

AC-04 Payment Decline

- Given the user enters invalid payment details
- When the payment gateway declines the transaction
- Then the system displays a clear payment error message to the user.

10. Agile Artifacts

10.1 User Stories

ID	User Story
US-01	As a customer, I want to view all items in my cart, so that I can review my order before checkout.
US-02	As a customer, I want to see the correct price, quantity, and totals in my cart, so that I can confirm the final amount before paying.
US-03	As a customer, I want the checkout button to reliably take me to the payment step, so that I can complete my purchase without interruption.
US-04	As a customer, I want to receive clear confirmation when my payment succeeds, so that I know my order has been placed.
US-05	As a customer, I want to receive a receipt by email after a successful purchase, so that I have a record of my transaction.

10.2 Business Rules

ID	Rule
BR-01	The system shall calculate and display cart subtotals and totals only when at least one item has been added to the cart.
BR-02	The system shall process a payment authorization request only when all required payment fields are provided.
BR-03	The system shall generate an order confirmation and send an electronic receipt only after payment authorization is approved.

11. Requirement Traceability

Problem Identified	Req ID	Mapped Requirement
Cart page fails to display all selected items	FR-01	The system shall display all items added to the cart by the user on the cart page.
Incorrect or missing pricing/totals during cart review	FR-03	The system shall calculate and display the per-item subtotal and the total cart amount.
Checkout button unresponsive / redirect failure	FR-07	The system shall redirect the user to the payment gateway interface upon checkout selection.
Users unable to complete payment due to unreliable authorization handling	FR-08	The system shall securely transmit submitted payment details to the payment gateway for authorization.
Users lack clarity on payment outcome (approved/declined)	FR-09	The system shall display the payment authorization status (approved/declined) returned by the payment gateway.
Users do not receive confirmation/receipt after successful purchase	FR-10	The system shall display an order confirmation and send an electronic receipt to the user's email upon payment approval; if declined, the system shall display a clear, actionable error message.

12. Success Metrics

- Reduce cart abandonment by up to 20%.
- Reduce customer churn by up to 10%.
- Reduce checkout-related support tickets by up to 20%.
- Checkout redirection completes within 3 seconds under normal operating conditions (NFR-01).